

# Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z4

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## Contents

1    Examined  $H(z)$  for CG TIA Automated NO L simple Z1 Z4:  $\frac{Z_1Z_4g_m}{2Z_1g_m+2}$

$$H(z) = \frac{Z_1Z_4g_m}{2Z_1g_m + 2}$$

2    AP

3    BP

4    BS

5    GE

6    HP

7    LP

7.1    LP-1  $Z(s) = \left( \frac{1}{C_1s}, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{R_4g_m}{2C_1C_4R_4s^2 + 2g_ms + (2C_1 + 2C_4R_4g_m)}$$

Parameters:

Q:  $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{g_m}}{C_1+C_4R_4g_m}$   
wo:  $\frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}}$   
bandwidth:  $\frac{C_1+C_4R_4g_m}{C_1C_4R_4}$   
K-LP:  $\frac{R_4}{2}$   
K-HP: 0  
K-BP: 0  
Qz: None  
Wz: None

7.2    LP-2  $Z(s) = \left( \frac{R_1}{C_1R_1s+1}, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{R_1R_4g_m}{2C_1C_4R_1R_4s^2 + 2R_1g_ms + s(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

Parameters:

Q:  $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}}{C_1R_1+C_4R_1R_4g_m+C_4R_4}$   
wo:  $\frac{\sqrt{R_1g_m+1}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$   
bandwidth:  $\frac{C_1R_1+C_4R_1R_4g_m+C_4R_4}{C_1C_4R_1R_4}$   
K-LP:  $\frac{R_1R_4g_m}{2R_1g_m+2}$   
K-HP: 0  
K-BP: 0  
Qz: None  
Wz: None

## 8 X-INVALID-NUMER

**8.1 X-INVALID-NUMER-1**  $Z(s) = \left( R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_4 g_m s + R_4 g_m}{2g_m + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m)}$$

**Parameters:**

Q:  $\frac{\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} g_m \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1} \sqrt{C_4} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}}}{C_1 R_1 g_m + C_1 + C_4 R_4 g_m}$   
 wo:  $\sqrt{\frac{g_m}{C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_4}}$   
 bandwidth:  $\frac{\sqrt{\frac{g_m}{C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_4}} (C_1 R_1 g_m + C_1 + C_4 R_4 g_m)}{\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} g_m \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1} \sqrt{C_4} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}}}$   
 K-LP:  $\frac{R_4}{2}$   
 K-HP: 0  
 K-BP:  $\frac{C_1 R_1 R_4 g_m}{2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m}$   
 Qz: None  
 Wz: None

## 9 X-INVALID-ORDER

**9.1 X-INVALID-ORDER-1**  $Z(s) = (R_1, \infty, \infty, R_4, \infty, \infty)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + 2}$$

**9.2 X-INVALID-ORDER-2**  $Z(s) = \left( R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 g_m}{s (2C_4 R_1 g_m + 2C_4)}$$

**9.3 X-INVALID-ORDER-3**  $Z(s) = \left( R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s (2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

**9.4 X-INVALID-ORDER-4**  $Z(s) = \left( R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{s (2C_4 R_1 g_m + 2C_4)}$$

**9.5 X-INVALID-ORDER-5**  $Z(s) = \left( \frac{1}{C_1 s}, \infty, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_4 g_m}{2C_1 s + 2g_m}$$

**9.6 X-INVALID-ORDER-6**  $Z(s) = \left( \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{g_m}{2C_1 C_4 s^2 + 2C_4 g_m s}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left( \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_4 g_m s + g_m}{2C_1 C_4 s^2 + 2C_4 g_m s}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left( \frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m}{2C_1 R_1 s + 2R_1 g_m + 2}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left( \frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 g_m}{2C_1 C_4 R_1 s^2 + s(2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left( \frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{2C_1 C_4 R_1 s^2 + s(2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left( R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_1 R_1 R_4 g_m s + R_4 g_m}{2g_m + s(2C_1 R_1 g_m + 2C_1)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left( R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_1 R_1 g_m s + g_m}{2C_4 g_m s + s^2(2C_1 C_4 R_1 g_m + 2C_1 C_4)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left( R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_4 R_1 R_4 g_m s^2 + g_m + s(C_1 R_1 g_m + C_4 R_4 g_m)}{2C_4 g_m s + s^2(2C_1 C_4 R_1 g_m + 2C_1 C_4)}$$

**10 X-INVALID-WZ**

**11 X-PolynomialError**